

The 60/40 verdict

Why a portfolio that worked for forty years failed in 2022, and what to do about it

ABSTRACT

We examine the historical performance of the 60/40 stock-bond portfolio across 152 years of monthly U.S. data. Across the full sample, the portfolio returned 7.69% annualised at 8.70% volatility with a maximum drawdown of 61.0% — modestly attractive risk-adjusted performance. The 1981–2021 sub-period, however, produced extraordinary results: 10.5% annualised at 9.8% volatility with maximum drawdown of approximately 30%. This forty-year era of attractive 60/40 performance was driven by two structural conditions — falling inflation and falling real interest rates — that are unlikely to recur. The 2022 calendar year (–16% on the 60/40, simultaneous double-digit losses on both legs) was not anomalous in the long-run record; it was the modal outcome in inflation regimes above 4%. We document the conditional behaviour of 60/40 across inflation regimes, examine alternatives that have historically performed better in inflationary episodes (commodities, gold, real assets, broad multi-asset risk-parity), and discuss what the 2022 experience implies for forward portfolio construction.

KEY FINDINGS

- ❖ **60/40's historical performance is decent, not extraordinary.** Full-sample (1871–2022): 7.69% annualised at 8.70% vol with –61.0% max drawdown. The legendary 1981–2021 era was an outlier.
- ❖ **The 1981–2021 era was not normal.** Falling inflation and falling real rates produced simultaneous tailwinds for both stocks and bonds. The two structural drivers were one-time, not recurring.
- ❖ **2022 was not exceptional in the long-run record.** Simultaneous double-digit losses on stocks and long bonds is the modal outcome in inflation regimes above 4%. The pattern was last observed 1973–80.
- ❖ **Real returns flip negative above 4% inflation.** 60/40 real annualised return: +12.7% in deflation, +7.0% in moderate inflation, –1.7% in elevated, –5.5% in high. The portfolio is regime-conditional.
- ❖ **The diversification answer is not 'less stocks/more bonds'.** Adding inflation-protective assets (commodities, gold, real assets, TIPS, multi-asset trend) provides genuine diversification across regimes. Reducing equity exposure does not.

EXEC Executive summary

The 60/40 portfolio is not dead. But it is also not what its forty-year track record suggests it is. The portfolio that performed extraordinarily from 1981 to 2021 was riding two structural tailwinds — falling inflation and falling real rates — that were each one-time and are now exhausted.

This note examines the long-run performance of the 60/40 stock-bond portfolio across 152 years of U.S. monthly data, with particular attention to its conditional behaviour across inflation regimes. The full-sample performance is modestly attractive: 7.69% annualised real total return at 8.70% volatility with a maximum drawdown of 61% (concentrated in the 1929–32 period). The Sharpe ratio of approximately 0.55 over the long run is competitive with most simple alternatives.

The 1981–2021 sub-period, however, produced returns and risk-adjusted ratios that materially exceeded historical norms. The 60/40 returned approximately 10.5% nominal annualised over this 40-year window, at volatility of approximately 10%, with a maximum drawdown of approximately 30%. This combination would imply a Sharpe ratio above 0.7 — well above the long-run norm. The era was, in retrospect, exceptional rather than typical.

The exceptional performance was not accidental. The 1981–2021 period coincided with two major structural conditions: a sustained decline in inflation (from approximately 14% in 1980 to approximately 1.5% in 2020) and a sustained decline in real interest rates (from approximately 8% to approximately –1%). The first produced strong equity tailwinds (PE expansion as inflation fell — see Paper 05). The second produced strong bond tailwinds (cumulative duration gains from secular rate decline). The two operated jointly, and the joint operation drove the portfolio's exceptional realised Sharpe.

The 2022 experience — simultaneous double-digit losses on both legs — was characterised in industry commentary as a "60/40 failure" or as an "unprecedented event." Neither characterisation is accurate. Simultaneous losses on stocks and long bonds is the modal outcome in inflation regimes above 4%, observed in 1916–20, 1946–48, 1973–80, and now 2022. The 1981–2021 era was not normal; the 2022 experience is not exceptional. The mistake was confusing the post-1981 sample with the long-run norm.

The implication for portfolio construction is that 60/40 is not a robust solution; it is a regime-conditional one. The portfolio works in low-inflation, falling-rate environments and fails in high-inflation, rising-rate ones. Genuine diversification across regimes requires assets that have historically held up in high-inflation regimes (commodities, gold, real estate, TIPS, multi-asset trend strategies). Reducing equity exposure or increasing bond exposure does not solve the problem; both of those legs have correlated inflation exposure.

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The note proceeds as follows. Section 2 documents the long-run record. Section 3 examines the 1981–2021 era specifically. Section 4 presents the conditional regime analysis. Section 5 discusses portfolio-construction alternatives.

01 The long-run record

We construct a 60/40 portfolio using Shiller's monthly U.S. equity total return series and an approximated long-duration government bond return series (using Shiller's "Long Interest Rate" series and a duration-adjusted price-impact methodology). The portfolio is monthly-rebalanced to maintain 60% equity / 40% bond weights. The sample runs from January 1871 through December 2022, providing 1,824 monthly observations.

TABLE 1. LONG-RUN PERFORMANCE OF 60/40 VS CONSTITUENTS, 1871–2022

Strategy	Ann. return	Ann. vol	Max drawdown	Sharpe
100% equity	+9.31%	14.01%	−81.8%	0.66
100% bonds	+4.63%	7.84%	−40.2%	0.59
60/40	+7.69%	8.70%	−61.0%	0.55

The 60/40 produces a slightly lower Sharpe ratio than either constituent in isolation. This is initially counterintuitive — diversification is conventionally expected to improve risk-adjusted returns — but is consistent with the empirical correlation structure. The long-run correlation between stocks and bonds in our sample is approximately +0.10 (positive but weak). At low correlation, the diversification benefit is modest, and the lower-return asset (bonds) drags overall return more than the volatility reduction compensates.

The maximum drawdown of 61% — concentrated in 1929–32, when both legs declined — is also worse than might be expected from a diversified portfolio. This reflects that the worst equity drawdowns in U.S. history have not always coincided with bond rallies; the 1929–32 period saw both equities and long bonds decline materially in real terms.

1.1 Decade-by-decade performance

TABLE 2. 60/40 RETURNS BY DECADE, REAL TERMS

Decade	60/40 real ann.	Equity real ann.	Bond real ann.
1870s	+8.4%	+7.8%	+8.6%
1900s	+5.2%	+10.3%	+1.6%
1910s	−4.8%	+4.5%	−9.8%
1930s	+1.6%	+0.0%	+5.0%
1950s	+11.3%	+19.3%	+0.4%
1970s	−2.5%	+5.8% (nom)	−4.5%
1980s	+12.2%	+17.3%	+8.7%
1990s	+11.6%	+18.0%	+5.7%
2000s	+1.4%	−0.7%	+5.4%
2010s	+8.7%	+13.3%	+1.6%
2020s †	−2.1%	+4.5%	−12.3%

† 2020s through 2022. Numbers are partial.

The decade-level pattern shows substantial variation. The 1870s and 1980s–2010s saw consistently positive 60/40 real returns. The 1910s, 1970s, and 2020s (so far) saw negative real returns, all three of which featured high inflation. The 1930s saw positive real returns despite the equity collapse, because deflation made the real value of both stocks and bonds positive.

02 The 1981–2021 era

The 40-year period from 1981 through 2021 was, for the 60/40 portfolio, the most favourable in the long-run sample. Annualised nominal returns were approximately 10.5% with realised volatility of approximately 10% — a Sharpe ratio of approximately 0.7, well above the long-run norm.

The exceptional performance was not random. It was driven by two specific structural conditions, each of which contributed independently to favourable returns on each leg:

Condition 1: Falling inflation. CPI inflation peaked at 14.6% year-over-year in March 1980 and declined to approximately 1.5% by 2020. The decline was secular and persistent, with brief reversals (1990, 2000, 2008) but a clear long-run trend. Falling inflation produced two effects on equities: it reduced the discount rate (raising present values) and produced PE expansion (CAPE rose from approximately 7 in 1981 to approximately 30 in 2021). Both effects were tailwinds for nominal equity returns.

Condition 2: Falling real interest rates. The 10-year real Treasury yield (approximated using the 10-year nominal yield minus contemporaneous CPI inflation) declined from approximately 8% in 1980 to approximately –1% in 2020. The 40-year decline produced cumulative duration-adjusted bond returns of approximately 200% above the equivalent constant-yield baseline. Bond holders received not just the coupon but also the price gain from the rate decline.

The two conditions operated jointly. Falling inflation contributed to falling nominal rates (via the inflation-expectations channel) and also to falling real rates (via reduced inflation risk premium). The compound effect on the 60/40 was multiplicative rather than additive: both legs received structural tailwinds from the same underlying secular shift.

TABLE 3. 1981–2021 VS FULL-SAMPLE COMPARISON

Metric	1981–2021	Full sample (1871–2022)
Equity ann. nominal	+11.8%	+9.3%
Bond ann. nominal	+8.1%	+4.6%
60/40 ann. nominal	+10.5%	+7.7%
60/40 ann. volatility	9.8%	8.7%
60/40 max drawdown	–30%	–61%
Implied Sharpe (gross)	~0.7	~0.55

The 1981–2021 Sharpe ratio of approximately 0.7 is materially higher than the full-sample 0.55. The difference is the structural tailwind: in this era, both legs benefited from sustained falling inflation and falling real rates simultaneously. The pattern was not coincidental and was not part of the long-run norm.

2.1 The exhaustion of the tailwinds

By 2020, both structural drivers had reached implausible end-points. Inflation had fallen from 14.6% to 1.5%; further decline at the same rate would imply persistent deflation, an outcome central banks would aggressively resist. Real interest rates had fallen from 8% to –1%; further decline would imply substantial negative real rates, which conventional monetary economics treats as unsustainable for extended periods.

The 2022 experience can be understood, in this framing, as the inevitable consequence of the tailwinds reaching their endpoints and reversing. Inflation rose from 1.5% to 9.1%, retracing approximately a third of the 1980–2020 decline in two years. Real rates rose from –1% to approximately +2%, retracing approximately a third of the 1980–2020 decline. Both legs of the 60/40 received the same shock in reverse: equities saw PE compression, bonds saw price losses from rising rates. The 60/40 returned approximately –16% in 2022, with both legs contributing to the loss.

The forward implication is that the 1981–2021 Sharpe of 0.7 is not the right anchor for forward portfolio expectations. The structural tailwinds that produced it are exhausted; the more relevant anchor is the long-run Sharpe of 0.55, which is consistent

with the unconditional historical norm and does not assume the recurrence of one-time secular shifts.

03 Conditional behaviour by inflation regime

The empirical conditioning of 60/40 returns on inflation regime is direct and dramatic. We classify each month in our sample by trailing 12-month CPI change and report 60/40 returns within each regime.

TABLE 4. 60/40 CONDITIONAL PERFORMANCE BY INFLATION REGIME, 1871–2022

Regime	N months	60/40 ann. nominal	60/40 ann. real
Deflation (<0%)	414	+8.2%	+12.7%
Low (0–2%)	414	+9.3%	+8.0%
Moderate (2–4%)	458	+10.8%	+7.0%
Elevated (4–6%)	190	+3.3%	–1.7%
High (>6%)	345	+3.5%	–5.5%

The conditioning is sharp. In deflation, low-inflation, and moderate-inflation regimes (cumulatively 71% of all months in the sample), the 60/40 produces real annualised returns of 7–13%. In elevated and high-inflation regimes (29% of months), real returns are –1.7% and –5.5% respectively.

The 2022 experience falls into the high-inflation regime. The realised –16% nominal (approximately –24% real after the year's 8% inflation) is, if anything, slightly worse than the regime average. The historical record shows that high-inflation regimes are bad for 60/40, and 2022 produced a bad outcome. The two are consistent.

3.1 Why the regime conditioning is sharp

The economic mechanism is straightforward. In high-inflation regimes:

- **Equities** face PE compression as inflation reduces the real present value of nominal cash flows (the "inflation illusion" mechanism of [Modigliani and Cohn, 1979](#)). Real earnings growth, the only positive source of real return, is typically below 2%.
- **Long bonds** face direct real-value erosion. The fixed nominal coupon is worth less in real terms as inflation rises; the duration-adjusted price impact of rising rates is also negative for long-duration positions.
- **Both effects** typically operate simultaneously, producing the simultaneous-loss pattern observed in 1916–20, 1946–48, 1973–80, and 2021–22.

The 60/40 portfolio is therefore not diversified against inflation regime — it is doubly exposed. The conventional wisdom that "stocks are inflation hedges" is empirically false above 4% inflation; the conventional wisdom that "bonds reduce risk" is empirically false during inflation regimes. A portfolio holding 60% of one asset that fails in high inflation and 40% of another asset that fails in high inflation is not diversified; it is concentrated.

04 Alternatives and conclusion

4.1 Genuine diversification

The historical alternatives that have provided genuine diversification across inflation regimes are:

- **Commodities.** Broad commodity baskets have historically produced positive real returns in high-inflation regimes ([Erb and Harvey, 2006](#); [Gorton and Rouwenhorst, 2006](#)). Caveat: post-2005 financialisation has eroded the inflation-hedge property somewhat.
- **Gold.** Long-run real returns on gold are positive in high-inflation regimes, though with substantial short-run noise. (See Paper 11 for detailed treatment.)
- **Real estate.** Direct real estate and equity REITs have historically held up in moderate inflation, with mixed results in high inflation. The 1970s saw strong real returns from real estate.

- **TIPS.** Treasury Inflation-Protected Securities provide explicit inflation linkage, though their availability is post-1997 only and their performance in deflationary regimes is asymmetric (the principal floor protects against deflation but not against rising real rates).
- **Multi-asset trend strategies.** Cross-asset trend-following strategies (see Paper 07 and Paper 09) have positive returns in inflationary regimes, particularly in fixed-income (where the rising-rate trend produces short-duration profits).

A portfolio holding 5–10% of each of two or three of these alternatives, funded from the 60/40 position, would have substantially smoothed the 2022 experience while producing modest opportunity cost in low-inflation regimes.

4.2 What does not solve the problem

Several proposed responses to 2022 do not actually address the underlying issue:

- **"Reduce equity, increase bonds."** Both legs of 60/40 have correlated inflation exposure. Shifting weight from equities to bonds increases inflation exposure, not decreases it.
- **"Reduce bonds, increase equities."** Symmetric problem. Increasing equity exposure does not diversify against inflation; equity real returns are negative in high-inflation regimes.
- **"Shorten bond duration."** Helps with rate volatility but not with inflation per se. Short-duration bonds in high-inflation regimes produce flat real returns at best.
- **"Wait for the regime to resolve."** Plausible but requires holding through extended drawdowns. Historical high-inflation regimes (1916–20, 1973–80) have lasted 4–7 years.

4.3 Conclusion

The 60/40 portfolio is not dead. Its long-run real return of approximately 5.5% (full-sample, after average inflation) is competitive with most simple alternatives. But the 1981–2021 era of exceptional performance, with Sharpe ratios above 0.7, was not normal — it was the consequence of two one-time structural tailwinds that have now reversed. The 2022 experience was not anomalous; it was the modal outcome in inflation regimes above 4%.

The path forward for portfolio construction is not to abandon 60/40 but to recognise its conditional nature. A portfolio that holds 60/40 as the core but allocates 10–20% to genuinely inflation-protective alternatives is meaningfully more robust across regimes than 60/40 alone. The cost of the alternatives is modest opportunity cost in low-inflation regimes; the benefit is materially reduced exposure to the regime that 60/40 cannot handle.

*The 60/40 of 1981–2021 is not the 60/40 of 1871–2022. The 1981–2021 Sharpe of 0.7 was a **tailwind**; the long-run Sharpe of 0.55 is the **norm**. The 2022 experience is not the failure of a working portfolio — it is the visibility of a portfolio's actual risk profile.*

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REF References

- Erb, C. B. and Harvey, C. R. (2006). The strategic and tactical value of commodity futures. *Financial Analysts Journal*, 62(2): 69–97.
- Fama, E. F. and Schwert, G. W. (1977). Asset returns and inflation. *Journal of Financial Economics*, 5(2): 115–146.
- Gorton, G. and Rouwenhorst, K. G. (2006). Facts and fantasies about commodity futures. *Financial Analysts Journal*, 62(2): 47–68.
- Ilmanen, A. (2012). *Expected Returns: An Investor's Guide to Harvesting Market Rewards*. Wiley.
- Leibowitz, M. L., Bova, A., and Hammond, P. B. (2010). *The Endowment Model of Investing: Return, Risk, and Diversification*. Wiley.
- Mehra, R. and Prescott, E. C. (1985). The equity premium: A puzzle. *Journal of Monetary Economics*, 15(2): 145–161.
- Modigliani, F. and Cohn, R. A. (1979). Inflation, rational valuation and the market. *Financial Analysts Journal*, 35(2): 24–44.
- Asness, C. S. (2022). The 60/40 portfolio is alive and well. *AQR Capital Management Perspectives*, October 2022.
- Bridgewater Associates (2022). *How to invest in a world of unanchored inflation*. Daily Observations, June 2022.
- Bank of America Global Research (2022). *The end of 60/40?* Cross-Asset Research, January 2022.
- Carl C. G. Eriksson · P. RealAlgos · The 60/40 verdict
- Vanguard Group (2022). *What 2022 means for the 60/40 portfolio*. Vanguard Research, December 2022.
- BlackRock Investment Institute (2023). *Resetting expectations: The new investment regime*. 2023 Outlook.

APX Appendix A: Sub-period Sharpe ratios

TABLE A1. 60/40 SHARPE RATIOS ACROSS 20-YEAR SUB-PERIODS

Period	60/40 ann.	Vol	Sharpe
1871–1890	+8.4%	8.2%	0.78
1891–1910	+7.5%	7.0%	0.71
1911–1930	+6.4%	9.4%	0.43
1931–1950	+5.8%	11.7%	0.31
1951–1970	+8.6%	7.6%	0.83
1971–1990	+8.9%	8.5%	0.75
1991–2010	+8.4%	8.2%	0.81
2011–2022	+6.7%	8.5%	0.51

The Sharpe ratio of 60/40 has varied between approximately 0.3 and 0.8 across 20-year sub-periods. The lowest Sharpe (0.31) occurred in 1931–1950, dominated by the Depression and WWII. The highest occurred in 1951–1970 (0.83) and 1991–2010 (0.81), both falling-inflation, falling-real-rate periods.

A.2 Stock-bond correlation through time

The realised stock-bond correlation has varied substantially. The full-sample correlation is +0.10 (positive but weak). The post-2000 correlation through 2020 was –0.30 (modestly negative — the period when bonds genuinely diversified equity risk). The post-2020 correlation is approximately +0.40 (strongly positive, the inflation-driven co-movement of 2022).

A.4 ProRealTime: 60/40 rebalancing snippet

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// 60/40 with quarterly rebalance
// Run on a synthetic portfolio basket – equity ETF + bond ETF

DEFPARAM CumulateOrders = True

equityWeight = 0.60
bondWeight = 0.40
rebalanceFreq = 63 // ~quarterly on daily bars

// At rebalance dates, compute current portfolio weights and adjust
IF BarIndex MOD rebalanceFreq = 0 THEN
    totalEquity = strategyprofit + capital
    targetEquity = totalEquity * equityWeight
    targetBond = totalEquity * bondWeight
    // Issue rebalancing orders to bring each leg back to target
    // (specifics depend on whether you run paired instances or a multi-instrument bot)
ENDIF
```

TABLE A2. STOCK-BOND CORRELATION BY SUB-PERIOD

Period	Correlation
1871–1900	+0.18

Period	Correlation
1900–1950	+0.05
1950–1980	+0.21
1980–2000	+0.08
2000–2020	–0.30
2020–2022	+0.42

The negative correlation period of 2000–2020 is anomalous in the long-run record. The post-2020 return to positive correlation is closer to the historical norm. Investors who calibrated 60/40 expectations against the 2000–2020 correlation are mis-calibrated against the broader history.

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